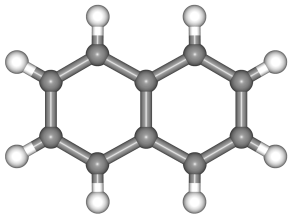
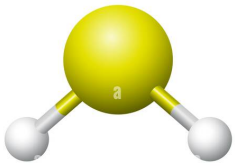
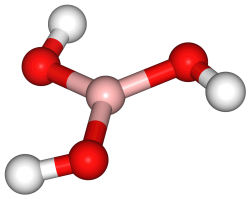
Inorganic Cumulative Exam
Sept 29 2023

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (8 pts) Draw the molecular orbital diagram of O_2.
   (b) (6 pts) Calculate the bond order of O_2, O_2^+, and O_2^{2+}.
   (c) (6 pts) Of the three oxygen molecules in (b), which has the longest bond length? Please explain your answer.
2. Identify the point group for each of the following molecules (4pts each). Please note any assumptions you make about the molecule when assigning the point group.

(a) benzene	(b) napthalene 	(c) H_2S 
(d) boric acid ($B(OH)_3$) 	(e) fluoroamine (NH_2F)	

3. The thiocyanate ion, SCN- is an example of an ambidentate ligand. (a) (8 pts) In a few sentences, describe what is meant by *ambidentate*, and how the ligand binds to a central atom. (b) (4 pts) Provide another example of an ambidentate ligand. (c) (8 pts) Give two examples of how the external environment can affect the binding of an ambidentate ligand such as SCN-
4. (a) (11 pts) Use the attached character tables to identify the irreducible representations of the vibrational modes of H_2S (structure shown in Q2). (b) (9 pts; 3 pts each) Sketch the motions of H_2S (using arrows) corresponding to each mode identified in (a).
5. (a) (10 pts) Explain the Pauling's electronegativity scale in terms of the HOMO/LUMO levels of atoms participating in a bond. (b) (10 pts) Explain why hydrogen halides form diatomic molecules while other elements from Group 1 form salts with halides.
6. Aluminum metal forms a face-centered cubic lattice, with a density of 2.7 g cm^{-3}. Estimate the Al-Al bond distance given the molar mass of Al as 26.98 g mol^{-1}.
7. Polonium metal is one of the only examples of a material with a simple cubic lattice. (a) (8 pts) Sketch the (110) planes for the cubic polonium lattice. (b) (12 pts) The lattice

constant for the simple cubic Po lattice is 3.36 Å. Assume you are using a powder diffractometer with a Cu Kα source (λ = 1.54 Å) to measure the diffraction pattern of a simple cubic Po powder. At what angle (2θ) would you measure the diffraction corresponding to the (110) planes of Po?

8. Transparent conducting oxides are ubiquitous in optoelectronic devices (solar cells, detectors) and consumer electronics (touch screens). (a) (8 pts) Describe how transparent conductive oxides are used in these devices. (b) (12 pts) Describe the electronic structure of a transparent oxide and now one such as SnO2 can be made conductive during its synthesis.
9. Assume that you are using an ionic liquid for an energy storage electrocatalytic process, such as electrochemical nitrogen reduction. Given the following list of properties, state if you would like each property to be "low" or "high" and explain your answer (5 pts each but must have an explanation to get any credit). (a) boiling point; (b) thermal stability; (c) electrical conductivity; (d) melting point
10. Assume you are required to prepare an extremely pure thin film of GaAs for an LED. (a) (9 pts) Provide a detailed synthetic plan, including any instrumentation, source material, or other consideration to make the film as pure as possible. (b) (5 pts) Why does this approach ensure high purity of the film? (c) (6 pts) Name two techniques that you will use to measure the purity of the film. *Note: while there are many answers to these questions, note the need for high purity. In that case, some answers will be more correct than others.*

Some useful character tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	-1	R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_2	E	$C_2(z)$	$C_2(y)$	$C_2(x)$		
A	1	1	1	1		x^2, y^2, z^2
B_1	1	1	-1	-1	z, R_z	xy
B_2	1	-1	1	-1	y, R_y	xz
B_3	1	-1	-1	1	x, R_x	yz

D_3	E	$2C_3$	$3C_2$		
A_1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	-1	z, R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

D_{2d}	E	$2S_4$	C_2	$2C_2'$	$2\sigma_d$		
A_1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1	z	xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_{3d}	E	$2C_3$	$3C_2$	i	$2S_6$	$3\sigma_d$		
A_{1g}	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	-1	1	1	-1	R_z	
E_g	2	-1	0	2	-1	0	(R_x, R_y)	$(x^2 - y^2, xy)$ (xz, yz)
A_{1u}	1	1	1	-1	-1	-1		
A_{2u}	1	1	-1	-1	-1	1	z	
E_u	2	-1	0	-2	1	0	(x, y)	

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	−1	−1	1	1	−1	−1	R_z	xy
B_{2g}	1	−1	1	−1	1	−1	1	−1	R_y	xz
B_{3g}	1	−1	−1	1	1	−1	−1	1	R_x	yz
A_u	1	1	1	1	−1	−1	−1	−1		
B_{1u}	1	1	−1	−1	−1	−1	1	1	z	
B_{2u}	1	−1	1	−1	−1	1	−1	1	y	
B_{3u}	1	−1	−1	1	−1	1	1	−1	x	

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2'	1	1	−1	1	1	−1	R_z	
E'	2	−1	0	2	−1	0	(x, y)	$(x^2 - y^2, xy)$
A_1''	1	1	1	−1	−1	−1		
A_2''	1	1	−1	−1	−1	1	z	
E''	2	−1	0	−2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	1	−1	−1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	1	−1	1	−1	1	1	−1		$x^2 - y^2$
B_{2g}	1	−1	1	−1	1	1	−1	1	−1	1		xy
E_g	2	0	−2	0	0	2	0	−2	0	0	(R_x, R_y)	(xz, yz)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	1	−1	−1	1	−1	−1	1		
B_{2u}	1	−1	1	−1	1	−1	1	−1	1	−1		
E_u	2	0	−2	0	0	−2	0	2	0	0	(x, y)	